

Medical Readmission Risk Assessment Report

Humber College Institute of Technology & Advanced Learning
Capstone Project - Computer Programming
Team SweatHogs: Gurmat Singh, Minh Nguyen, Yuvraj Patel, Robert Johnson

Generated: July 14, 2025 at 04:41 PM

Patient Information

Age Group	[50-60)	Previous Inpatient Visits	1
Gender	Female	Previous Outpatient Visits	2
Time in Hospital (days)	5	Previous Emergency Visits	0
Admission Type	1	Diabetes Medication	Yes
Discharge Disposition	1	Medication Change	Ch
Admission Source	1	A1C Test Result	Norm
Number of Medications	10	Max Glucose Serum	None
Lab Procedures	30	Insulin	No
Medical Procedures	2	Metformin	Steady
Number of Diagnoses	3	Primary Diagnosis	250.00

Risk Assessment Results

Readmission Risk Score: 0.743
Risk Level: **High Risk**
Model Confidence: 74.3%

Chart generation error: expected str, bytes or os.PathLike object, not ImageReader

AI-Generated Medical Insights

Despite well-controlled diabetes (normal A1C, metformin steady, no insulin, primary diagnosis 250.00), this female patient in her 50s has a high readmission risk score of 0.74. This suggests the elevated risk is likely driven by her other two diagnoses, the acute reason for her emergency admission, or the complexity of her overall care (10 medications, regimen changed, 30 lab procedures).

****Insight/Remedy:**** Focus on robust post-discharge planning, including thorough medication reconciliation and close, multidisciplinary follow-up for all her medical conditions, not just diabetes, to mitigate this elevated readmission risk. Patient education on new medications and symptom monitoring is crucial.

****Disclaimer:**** This information is for general insight and should not be considered medical advice. Always consult with a qualified healthcare professional for diagnosis and treatment.

Understanding Your Risk Score

The readmission risk score is calculated using advanced machine learning algorithms that analyze multiple patient factors including medical history, current medications, diagnosis codes, hospital stay characteristics, and previous healthcare utilization patterns.

Risk Categories:

- **Low Risk (0.0-0.3):** Minimal likelihood of readmission
- **Medium Risk (0.3-0.7):** Moderate readmission risk requiring attention
- **High Risk (0.7-1.0):** Elevated readmission risk requiring intensive follow-up

IMPORTANT MEDICAL DISCLAIMER:

This report is generated by an AI system for informational purposes only and should not be considered medical advice, diagnosis, or treatment recommendations. Always consult with qualified healthcare professionals for medical decisions. The predictions are based on statistical models and may not account for all individual patient factors.

Generated by SweatHogs Medical AI Prediction System
Humber College Capstone Project | Computer Programming Program
Team Members: Gurmat Singh, Minh Nguyen, Yuvraj Patel, Robert Johnson